Separating definition from denominator in NHANES diabetes prevalence estimates: a cross-sectional, design-based comparison of HbA1c and fasting glucose surveillance definitions

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Abstract

Background. Diabetes prevalence estimated from the National Health and Nutrition Examination Survey (NHANES) depends on the biomarker definition, but each biomarker is measured in its own laboratory sample with its own weight. A difference between glycated hemoglobin (HbA1c)-based and glucose-based estimates can reflect the definition, the sample, or both. This study separated the two for the August 2021 to August 2023 release, the first with a phlebotomy weight.

Methods. Nonpregnant adults aged 20 and older were classified by HbA1c ≥ 6.5%, fasting plasma glucose ≥ 126 mg/dL, and self-reported diagnosis. Eight prespecified configurations crossed phenotype, laboratory domain, and weight. The primary contrast compared HbA1c-defined and glucose-defined prevalence in one paired domain (n = 2,938, fasting subsample weight) with a design-based paired difference. The secondary contrast compared HbA1c in that domain with HbA1c in its laboratory domain (n = 5,729, phlebotomy weight) by joint linearization. Sensitivity analyses, a 2017 to March 2020 replication, and reconciliation with published estimates were prespecified.

Results. In the paired domain, 10.8% (95% CI 9.1 to 12.7) met the HbA1c threshold and 11.2% (9.5 to 13.1) the glucose threshold, a paired difference of −0.41 percentage points (−1.15 to +0.33); adults without a reported diagnosis contributed −1.12 points (−1.69 to −0.56) to that full-population difference (not prespecified). Laboratory-domain HbA1c prevalence differed from paired-domain HbA1c by −0.24 points (−1.44 to +0.96). Agreement was 95.0%, yet 37% of adults above either threshold exceeded only one. Of adults reporting diagnosed diabetes, 20.3% had both biomarkers below threshold. The surveillance composite (15.8%) and its parts reproduced all 18 published national point estimates. In 2017 to March 2020 the sample and weight part was +0.07 points (−1.32 to +1.47) and the definition part −1.54 points (−2.71 to −0.37).

Conclusions. In both releases HbA1c estimates in the laboratory and paired domains differed by under a quarter of a point (95% intervals within 1.5 points of zero), while the tests identified partly different adults. Definitions should be compared in one paired domain with one weight and a covariance-aware difference, and estimates should name their denominator and weight. A checker that refuses incompatible combinations is provided.

Keywords: NHANES; diabetes; glycated hemoglobin; fasting plasma glucose; survey weights; complex survey design; surveillance definitions; measurement; reproducibility

Background

The National Health and Nutrition Examination Survey (NHANES) is the main source of national estimates of undiagnosed diabetes in the United States because it pairs an interview with laboratory measurement [1, 2]. Surveillance definitions combine a self-reported diagnosis with one or both of two biochemical thresholds [1, 2] drawn from clinical criteria: glycated

hemoglobin (HbA1c) at or above 6.5% [3–5] and fasting plasma glucose at or above 126 mg/dL [3]. The two tests reflect different aspects of glycemia, and their cut points classify partly different people [3, 4].

Several NHANES studies have compared the definitions. Carson and colleagues found lower undiagnosed prevalence with HbA1c than with fasting glucose in adults who attended a morning examination [6]. Cowie and colleagues estimated HbA1c-defined diabetes in the examined sample and compared definitions in a smaller oral glucose tolerance subsample, noting that some estimates “may differ slightly” between the two samples [7]. Davidson and Pan compared fasting glucose and HbA1c criteria within the fasting sample [8], Menke and colleagues reported national prevalence of 12.3% with HbA1c or fasting glucose and 14.3% when the 2-hour glucose criterion was added [9], and Selvin, Fang, and colleagues showed that a confirmatory definition requiring both tests changes undiagnosed estimates substantially [10, 11]. In a rural South African cohort, HbA1c-defined prevalence was about four times the fasting glucose-defined prevalence (kappa 0.23) [12].

These comparisons share a design problem that is easy to miss. In NHANES, HbA1c is measured in all examined participants who give blood, while fasting glucose is measured only in a morning fasting subsample. Each has its own weight. An analyst who takes HbA1c from its larger sample and glucose from the fasting subsample compares two definitions and two denominators at the same time. Most published analyses restrict both tests to the fasting subsample [1, 6, 8, 11], which isolates the definition but leaves the size of the sample effect unknown. Others describe their weights only in general terms or combine weights across components [9, 13].

The August 2021 to August 2023 release makes the question sharper. The National Center for Health Statistics (NCHS) added a phlebotomy weight (WTPH2YR) to address nonresponse to the blood draw and advises its use for blood analytes, while fasting glucose keeps its own subsample weight (WTSAF2YR) [14–17]. The release also had lower response than earlier cycles [15]. NCHS guidance published in 2026 asks analysts to define the denominators of proportions and to name the weights they use [18]. Recent analyses that include this release restrict HbA1c to the paired fasting sample [1, 19], compare people without fasting glucose across pooled cycles without weights [20], or leave the weight for a mixed definition unclear [13]. Two federal products report different total prevalence for the same release, partly because they differ in age range and weighting [1, 2].

This study asked how much of the difference between HbA1c-based and fasting glucose-based prevalence in the August 2021 to August 2023 release comes from the definition and how much from the laboratory sample and weight. It estimates both parts with design-based uncertainty, repeats the analysis in the 2017 to March 2020 release, and provides a reusable checker that refuses invalid comparisons. The audit updates earlier comparisons by separating the definition difference from the combined sample and weight difference.

Methods

Use of artificial intelligence (AI) tools

I designed the study, wrote and reviewed the code, performed and checked the analyses, and reviewed the results, references and manuscript. Claude Code (Anthropic) and ChatGPT (OpenAI) assisted with coding, data analysis, literature searching and manuscript preparation. Codex (OpenAI) assisted with code and result checks, editorial revision and document

formatting. Exact model versions were not recorded. I checked the AI-assisted code, results and references as described under Verification. I take full responsibility for the work.

Data source

The primary data were the NHANES August 2021 to August 2023 public-use files for demographics (DEMO_L), glycohemoglobin (GHB_L), plasma fasting glucose (GLU_L), the fasting questionnaire (FASTQX_L), the diabetes questionnaire (DIQ_L), and reproductive health (RHQ_L) [14, 16, 17, 21–24]. Files were downloaded on 2026-10-07 by following the links on the official NCHS listing pages, checked against the SAS transport file header, and recorded with Secure Hash Algorithm 256-bit (SHA-256) checksums that the build recomputes before every run. The release has 15 masked variance strata and 30 masked primary sampling units (PSUs), giving 15 design degrees of freedom. No newer complete release with these components was available on the access date. The NHANES 2017 to March 2020 pre-pandemic files (P_DEMO, P_GHB, P_GLU, P_FASTQX, P_DIQ, P_RHQ; 24 strata, 49 PSUs) were analyzed separately as a replication, with their own weights [25–30]. The two releases were not pooled and no trend was estimated, following NCHS guidance to use caution when combining August 2021 to August 2023 data with earlier cycles [15].

Analysis plan

The analysis plan was written and frozen, with a recorded SHA-256 hash, before any threshold classification or survey estimate was computed. It was an internal, timestamped plan and was not registered with an external registry. It fixed the domains, phenotypes, eight configurations, contrasts, sensitivity analyses, suppression rule, and a qualitative description of a useful negative

result. Before freezing it, I had read the NCHS documentation, whose codebooks print unweighted frequencies, and the published NCHS estimates for the release.

Additional file 1 records eight dated amendments. They cover two narrow-domain contrasts already described under sensitivity analysis 1, a per-release mapping of weight variables needed for the replication, the details of the NCHS reconciliation and of the interval method, three reporting notes (suppressed group components, a treatment-unknown row, and item nonresponse above 5% reported as a limitation), a note that the prior-work log and the contribution statement were finalized after the first estimation run, and three analyses that were not prespecified (described below). None changed a phenotype definition, threshold, domain, 2021 to 2023 weight, or one of the eight configurations.

Analysis domains and weights

Each participant in the demographics file contributed one record to the survey design. Domains were formed by subsetting that design, retaining all PSUs. The adult domain (A) included interviewed and examined participants aged 20 or older with a usable answer to the diagnosis question. Pregnancy was resolved with the examination variable RIDEXPRG (positive test or self-report at the examination), whose public universe is women aged 20 to 44. Participants coded as pregnant were excluded, as were those whose status could not be ascertained at the examination but who reported a current pregnancy at the interview (RHD143). Women whose status could not be ascertained and who did not report a pregnancy were retained (27 women in domain A in 2021 to 2023 and 58 in 2017 to March 2020). For men and for women outside the 20 to 44 universe the item was recorded as not applicable, not as a negative result. The diagnosis question asks women aged 20 or older about diabetes “other than during pregnancy”, so

gestational-only diagnoses are not coded as diabetes. The gestational diabetes history question (RHQ162 in the 2017 to March 2020 file) was dropped in this release [24, 30], so that history was unavailable.

Four laboratory domains were nested in A (Table 1). The HbA1c laboratory domain (H) required a positive phlebotomy weight and an HbA1c result and used WTPH2YR. The fasting glucose domain (F) required a positive fasting subsample weight, which NCHS assigns only to subsample members who gave blood after a fast of 8 to less than 24 hours, and a glucose result; it used WTSAF2YR. The common paired domain (P) was F restricted to participants with an HbA1c result and also used WTSAF2YR. The general examination weight was used only to describe who lacked laboratory data in A. In the 2017 to March 2020 release, which has no phlebotomy weight, domain H used the examination weight WTMECPRP, following the HbA1c documentation's instruction to use the sample weights in the demographics file [26].

Phenotypes

Thresholds followed the American Diabetes Association criteria [3] with inclusive boundaries: HbA1c at or above 6.5% (48 mmol/mol; LBXGH, reported to one decimal) and fasting plasma glucose at or above 126 mg/dL (7.0 mmol/L; LBXGLU, whole mg/dL). Diagnosed diabetes was a "yes" to DIQ010; "borderline" and "no" were coded as not diagnosed. Composite definitions were biochemical positivity on either test, the surveillance composite of diagnosis or either test, and diagnosis or HbA1c alone. A single elevated measurement in a survey is not a clinical diagnosis, which requires confirmation [3], so all results are estimates under surveillance definitions. The full specification, with units, codes, and boundaries, is version 1.0.0 of the phenotype specification in Additional file 2.

Configurations and contrasts

The eight configurations, K1 to K8, crossed phenotype, domain, and weight (Table 1). Table 1 also lists three estimates that the plan declared for other purposes: AUX1, needed only for the composite decomposition, and K1_S1 and K2_S1, the sensitivity analysis 1 estimates in the narrow fasting domain. The primary contrast was HbA1c-defined minus glucose-defined prevalence in P (K1 minus K2). Both proportions were estimated in one call so that their covariance entered the standard error of the difference, and the difference was tested with a design-based t statistic. No independent-samples test was used.

The secondary contrast was HbA1c in H minus HbA1c in P (K3 minus K1). It compares two estimators of nominally the same population quantity that differ in sample and weight, and it is described as a combined selection and weighting comparison, not a causal effect. Because the two estimators use different weights and overlapping records, each was written as a ratio of weighted totals, all four totals were linearized jointly on a unit-weight design with the same strata and PSUs, and the variance of the difference came from the delta method. The naive cross-domain difference, HbA1c in H minus glucose in P, equals the sum of these two contrasts exactly. The same decomposition was applied to the composite with and without glucose.

Estimation

Estimates used Taylor series linearization with strata and PSUs (nest = TRUE) in the R survey package [31, 32]. Proportion intervals were Korn-Graubard intervals with the degrees-of-freedom adjustment [33, 34]; logit intervals were computed alongside to check behavior near zero. Contrast intervals used t quantiles with the degrees of freedom of the PSUs containing domain members. All percentages are weighted survey estimates unless labeled unweighted.

Each proportion was classified as presented, flagged, or suppressed by the NCHS Data Presentation Standards for Proportions [34]. Suppressed estimates are not shown as numbers in the display tables, are hatched in Figure 1, and are not interpreted. Their values remain in the aggregate files, marked as suppressed, for audit.

Agreement between the threshold classifications in P was summarized by the weighted two-by-two table, percent agreement, and Cohen's kappa with a delta-method standard error. Missing laboratory data in A were described by age, sex, and race and Hispanic origin with the examination weight. Race and Hispanic origin is a self-reported social classification and was not interpreted as a biological mechanism. Supplementary estimates gave the paired difference by the same groups. Fasting glucose was never imputed from HbA1c, and no ad hoc reweighting was done.

Sensitivity analyses

Three sensitivity analyses were prespecified. First, a narrow fasting rule required, in addition to the NCHS 8 to less than 24 hour rule, that none of the intake items the fasting questionnaire asks analysts to consider (coffee or tea with cream or sugar, alcohol, gum or mints, antacids, supplements) was taken within 8 hours of the blood draw. The narrow domain kept WTSAF2YR and is a subdomain, not a reweighted sample. The 311 participants of all ages (245 of them adults in domain A) whose glucose values carry a zero fasting weight because they did not meet the 8 to less than 24 hour fasting criterion, including 2 with no recorded fasting time, were not added to any domain. Second, each biomarker was estimated in its own single-biomarker domain and compared with P. Third, diagnosed diabetes, biochemical positivity, and the surveillance

composite were compared to show how a definition that ignores diagnosis treats adults whose treatment may have lowered their values.

Analyses not prespecified

Three analyses were added after the planned analyses (amendment 8) and are labeled as not prespecified wherever they appear. First, the primary contrast was split into its parts among adults without and with a reported diagnosis; each part is a share of all adults in P, and the two parts sum to the primary contrast exactly. Second, agreement between the thresholds was summarized among adults in P without a reported diagnosis, the group studied in earlier comparisons [6]. Third, the primary contrast was repeated with the HbA1c cut point one reporting unit (0.1 percentage units) below and above 6.5%, to show how sensitive it is to small calibration differences. All three used the same domain, weight, and paired estimation as the primary contrast (Additional file 1, Table S7).

Configuration checker

The domain and weight rules were encoded in a configuration file and enforced by a checker that runs before estimation. It refuses fasting glucose without the fasting weight, a phenotype that needs a measure its domain does not guarantee, a weight that does not belong to the domain, a definition contrast between different domains or weights, glucose imputed from HbA1c, and pooled releases (Additional file 1, Table S1). A refusal is the intended result for these requests.

Verification

Three kinds of automated checks were kept apart. Software checks confirmed join cardinality and that domains nested as declared. Python code written separately from the R estimation

reproduced every reported point estimate from direct weighted sums, and an independent Python implementation of the linearized variance reproduced the R standard errors of every proportion and contrast and the covariance of the two primary estimates; the delta-method standard errors of kappa and the agreement ratios were not recomputed. Source reconciliation compared counts in the data files with the frequencies printed in the NCHS codebooks (91 codebook rows for 2021 to 2023 and 85 for 2017 to March 2020). External reconciliation compared estimates with NCHS Data Brief 516, which used the same age range and thresholds, excluded pregnant women, and used fasting sample weights [1]; sample sizes matched in all 18 published rows. All 39 automated checks passed (Additional file 1, Table S2). These checks test the software and its agreement with official sources; they are not independent expert review of the conclusions.

I repeated key steps by hand, outside the pipeline. After downloading the six 2021 to 2023 component files again from the NCHS listing pages, I confirmed that their SHA-256 checksums matched the source manifest and compared each variable name, response code, universe, and weight instruction in the phenotype specification and data dictionary with the NCHS documentation pages. I reran the full pipeline on my computer from a clean extraction of the source archive, and the rerun outputs matched the reported results within the declared tolerance.

For 25 randomly selected participants in domain P, I looked up the raw HbA1c, glucose, diagnosis, pregnancy, and weight values and applied the written definitions by hand. Every classification agreed with the pipeline. I recomputed the size of domain P and the numerators and denominators of the two primary estimates directly from the raw files with a separate short script. I compared the composite, diagnosed, and undiagnosed estimates with the published NCHS tables, checked every reference against its digital object identifier (DOI), PubMed record, or source web page, and read every table and figure against the result files.

Software and reproducibility

Python 3.13.5 with DuckDB 1.5.6 [35] and pandas 3.0.6 read the files, ran the Structured Query Language (SQL) queries that build the analytic frame and phenotypes, and produced figures and exports; R 4.5.1 (2025-06-13) with survey 4.5 produced every estimate [31, 32]; documents were rendered with Quarto 1.6.42 (bundled pandoc and typst). Dependencies are pinned (requirements.lock, renv.lock). Aggregate results are provided in Additional file 3. Reporting follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement for cross-sectional studies where it applies (Additional file 4) [36]. A full rerun from cached inputs reproduced every reported table value within an absolute tolerance of 1e-9. Run identifier: 20261007T212439Z.

Results

Participants and domains

Of 11,933 participants in the 2021 to 2023 demographics file, 8,860 were examined and 6,064 were examined adults aged 20 or older. After excluding 41 women who were pregnant at the examination and 1 who reported a pregnancy at the interview, domain A included 6,022 adults (no adult lacked a usable diagnosis answer). Domain H included 5,729 adults; 268 gave no blood and 25 had a positive weight but no HbA1c result. Domain F included 2,940 adults: 2,626 adults in A were not assigned to the fasting subsample, 211 were assigned but gave no blood, and 245 did not meet the fasting criterion. Domain P included 2,938 adults, which equals the sample size NCHS reports for this release [1]; the full participant flow for both releases is in Additional file 1, Table S3. Most of the missing fasting glucose was missing by design: among adults in A (weighted with WTMEC2YR), 43.7% were not assigned to the subsample and 48.9% had a valid

glucose value (Figure 1; Additional file 1, Table S4). Non-Hispanic Black adults had a lower estimated share with a valid fasting glucose (41.9%, 95% confidence interval [CI] 34.8 to 49.3, versus 48.9% overall), an interval that includes the overall value, and a higher share lacking HbA1c (8.3%, 6.0 to 11.0, versus 4.9%, 4.0 to 5.8).

The same adults, two thresholds

In P, 10.8% of adults (95% CI 9.1 to 12.7; 371 of 2,938 unweighted) had HbA1c at or above 6.5% and 11.2% (9.5 to 13.1; 383 of 2,938) had fasting glucose at or above 126 mg/dL (Figure 2, Table 1). The paired difference was −0.41 percentage points (−1.15 to +0.33; p = 0.25; Table 2). The sampling correlation between the two estimates was 0.91, so a standard error that ignored the covariance (1.17 points) would have been 3.4 times the paired standard error (0.35 points).

Similar prevalence did not mean the tests found the same people (Table 3). Both thresholds were met by 8.5% of adults, HbA1c only by 2.3%, and glucose only by 2.7%. Percent agreement was 95.0% (93.8 to 96.1) and kappa was 0.74 (0.70 to 0.79). Of adults above either threshold, 17.0% were above the HbA1c threshold only and 20.1% were above the glucose threshold only.

In analyses that were not prespecified (Additional file 1, Table S7), the paired difference was split by diagnosis status. Among adults without a reported diagnosis, fewer met the HbA1c threshold than the glucose threshold (2.6% versus 3.7% of all adults in P; −1.12 points, −1.69 to −0.56), whereas among adults with a reported diagnosis more met the HbA1c threshold (8.2% versus 7.5%; +0.71 points, +0.20 to +1.22). The two parts sum to the paired difference. Among adults without a reported diagnosis, kappa was 0.55 (0.49 to 0.61), and 61% (55 to 66) of those above either threshold were above only one.

HbA1c in its laboratory sample

In domain H, with the phlebotomy weight, 10.5% (9.2 to 12.0; n = 5,729) had HbA1c at or above 6.5%. The selection and weighting difference from the paired-domain estimate was −0.24 points (−1.44 to +0.96; p = 0.68). The naive comparison of HbA1c in H with glucose in P gave −0.65 points (−2.02 to +0.72), which equals the definition part (−0.41) plus the sample and weight part (−0.24) exactly (Figure 3, Table 2). The larger domain gave a more precise HbA1c estimate (standard error 0.62 versus 0.82 points), and its difference from the paired-domain estimate was small relative to that uncertainty.

For the composite, diagnosis or HbA1c in H was 14.0% (12.5 to 15.7), compared with 15.8% (13.6 to 18.2) for diagnosis, HbA1c, or glucose in P. Of this −1.75 point difference, −1.92 points came from leaving out glucose (paired, −2.48 to −1.37) and +0.17 from the change of sample and weight (−1.31 to +1.66).

Diagnosis, composites, and the official estimates

Diagnosed diabetes was 11.3% (9.3 to 13.5) in P. Adding biochemical positivity raised the composite to 15.8%; undiagnosed biochemical positivity was 4.5% of adults (95% CI 3.3 to 5.9; Additional file 1, Table S5). Undiagnosed adults above both thresholds, a definition close to the confirmed definition of Selvin and Fang [10, 11], made up 1.8% (1.3 to 2.4) of adults (Figure 4). A biochemical definition that ignored diagnosis gave 13.5% (11.4 to 15.8); the composite was +2.30 points (+1.64 to +2.95) higher. Among adults reporting diagnosed diabetes, 20.3% (16.0 to 25.3) had both biomarkers below threshold, as did 16.3% (12.3 to 20.9) of those reporting current insulin or diabetes pills. The estimate for diagnosed adults reporting no treatment was suppressed under the NCHS standards.

The composite, diagnosed, and undiagnosed estimates matched NCHS Data Brief 516 in all 18 published rows (overall, by sex, and by three age groups) for sample size, percent, and standard error to the published rounding (Additional file 1, Table S5). Korn-Graubard intervals matched in 17 of 18 rows. For diagnosed diabetes at ages 40 to 59, the interval here was 9.8 to 14.7 and the published interval 10.0 to 14.5. The difference came from capping the effective sample size at the actual sample size, as the NCHS presentation standards specify [34]; without the cap the interval was 10.0 to 14.5, matching the publication.

Sensitivity analyses

The narrow fasting rule removed 172 adults from P, mainly for supplements (93), gum or mints (59), and antacids (31) taken within 8 hours; an adult could fail more than one item. In the narrow domain the HbA1c and glucose estimates were 10.4% and 10.9%, and the paired difference was −0.50 points (−1.32 to +0.33) (Figure 5). Glucose in its single-biomarker domain F (n = 2,940) was 11.2%, −0.01 points from the paired-domain value, because only 2 adults in F lacked HbA1c. The third sensitivity analysis is reported in the previous section. In an analysis that was not prespecified, moving the HbA1c cut point by one reporting unit changed the paired difference to +0.85 points (−0.10 to +1.80) at 6.4% and −1.30 points (−2.01 to −0.59) at 6.6%, because 1.3% of adults in P had an HbA1c of exactly 6.4% and 0.9% exactly 6.5%. In 2017 to March 2020 the corresponding differences were −0.33 (−1.51 to +0.85) and −2.35 points (−3.61 to −1.10).

Groups

The paired difference ranged from −1.1 to 0.0 points across age and sex groups, with every interval between −3.5 and +1.5 points (Additional file 1, Table S6 and Figure S1). For race and

Hispanic origin groups, 2 component estimates (non-Hispanic Asian and non-Hispanic Black adults) were suppressed under the NCHS standards; those group differences are shown in Additional file 1 for completeness and are not interpreted.

Replication in 2017 to March 2020

The replication used 3,757 adults in P, again equal to the NCHS sample size for that release [1], and 8,000 in H. HbA1c and glucose prevalence in P were 10.3% (8.4 to 12.4) and 11.8% (10.3 to 13.5), a paired difference of −1.54 points (−2.71 to −0.37; p = 0.01). HbA1c in H with the examination weight was 10.4% (9.4 to 11.5), a sample and weight difference of +0.07 points (−1.32 to +1.47). Kappa was 0.71, and 19.3% of diagnosed adults had both biomarkers below threshold. Split by diagnosis status (not prespecified), the part among adults without a reported diagnosis was −1.48 points (−2.18 to −0.78) and the part among adults with one −0.07 points (−1.10 to +0.96); kappa among adults without a reported diagnosis was 0.52 (0.44 to 0.60). In this release 451 of 8,451 adults in A (5.3% unweighted; 4.2% weighted, 95% CI 3.4 to 5.1) lacked HbA1c, and the examination weight does not adjust for that nonresponse.

Discussion

In both releases the estimated sample and weight part was small: −0.24 points (−1.44 to +0.96) in 2021 to 2023 and +0.07 points (−1.32 to +1.47) in 2017 to March 2020. Both intervals include zero and also allow shifts of up to about 1.5 points, so the data exclude only differences larger than about 1.5 points rather than show that there is none. The plan described this outcome in advance, without a numerical margin, as a useful negative result. In 2017 to March 2020 the point estimates place most of the naive cross-domain difference in the definition part, although the sample and weight interval allows a part nearly as large; in 2021 to 2023 neither part could

be distinguished from zero. The definition part was larger in 2017 to March 2020 than in 2021 to 2023. The releases were not compared formally, and the difference may reflect sampling error, the HbA1c instrument change, or lower response, so no trend is implied. Similar totals still concealed disagreement about individuals: an estimated 37% (95% CI 31 to 43) of adults above either threshold were above only one.

These results agree with earlier NHANES work in direction. Carson and colleagues found lower undiagnosed prevalence with HbA1c than with fasting glucose and a kappa of 0.60 among adults without diagnosed diabetes [6]; in the same group here, HbA1c identified fewer adults than glucose in both releases (−1.12 and −1.48 points of all adults in P), with kappa of 0.55 and 0.52. Davidson and Pan reported higher prevalence with the glucose criterion than with HbA1c in 1999 to 2010 (11.5% versus 10.5%) [8]. The confirmatory definition of Selvin and Fang lowers undiagnosed estimates [10, 11], as the both-tests cell did here. In 2021 to 2023 the small all-adult difference is the net of this negative gap and a positive gap among adults with a reported diagnosis (+0.71 points), many of whom were treated; these splits were not prespecified. Poor concordance in a South African cohort [12] and documented differences in HbA1c performance across populations [37] mean that the small national difference here should not be read as agreement within every group.

The contribution of this study is the separation of the two parts and their uncertainty. Cowie and colleagues noted that estimates from the examined sample and from a subsample could differ [7], and Hu and colleagues reported an unweighted comparison of people without fasting glucose that pooled design-based subsampling with nonresponse [20]. Here the sample and weight part is estimated with the documented weight for each domain and with its covariance, so an analyst can see that, in these releases, the HbA1c laboratory domain and the paired fasting domain gave

similar HbA1c prevalence within the precision available. The distinction also matters when official numbers are compared. NCHS reported total prevalence of 15.8% for adults 20 and older with one fasting weight [1], and this audit reproduced it. The National Diabetes Statistics Report gives 14.9% (13.1 to 16.9) for adults 18 and older [2]. Its documentation describes a different age range, a different handling of adults with a missing laboratory value, and component-specific weights [2]; this audit did not reproduce that estimate and did not measure how much each difference contributes.

Ignoring covariance would have increased the standard error 3.4-fold. This illustrates why definition comparisons need one paired domain, one weight, and a covariance-aware difference. When HbA1c is analyzed alone, its laboratory domain with the phlebotomy weight gives a more precise estimate that agreed with the paired-domain value within its uncertainty. Diagnosed diabetes also needs to remain separate from biochemical positivity. In this analysis, 20.3% of diagnosed adults, and 16.3% of those reporting insulin or diabetes pills, had both values below threshold, so biochemical definitions miss many treated people. Reporting the denominator, weight, and sample universe for each estimate makes the comparison explicit, as NCHS guidance asks [18]. The configuration checker enforces these requirements before estimation.

Limitations

Each test was measured once, so positivity on a single test is not a confirmed diagnosis [3, 38]. Positivity on both tests in the same sample meets the American Diabetes Association laboratory confirmation rule [3], but no participant was clinically evaluated. Diagnosis was self-reported, and self-reports of diabetes can be inconsistent over time [39], so some misclassification is likely. The borderline category was coded as not diagnosed.

The primary HbA1c instrument changed from the Tosoh G8 to the Bio-Rad D-100 during 2021 to 2023, and the secondary method used to retest 1.9% of specimens also changed instruments. The public file has no participant-level indicator or changeover date. In the NCHS bridging study, Tosoh G8 values averaged 2.3% higher in relative terms than Bio-Rad D-100 values, within the 3% total allowable error, and NCHS applied no adjustment [16]. Because HbA1c is reported to one decimal, a calibration difference of 0.1 percentage units would move the paired difference by about 0.9 to 1.3 points, more than its interval half-width, so the size and sign of the 2021 to 2023 definition difference depend on a calibration that the public files cannot check.

The unweighted examination response rate was 25.7% in 2021 to 2023, compared with 46.9% in 2017 to March 2020 [15]. The weights adjust for nonresponse but cannot remove all bias, and NCHS has not yet published the full weighting report, so the relationship between WTSAF2YR and WTPH2YR is undocumented. The sample and weight part is descriptive: it measures how two documented estimators differ, not what would happen under a different design. Its variance, like all variances here, treats the published weights as fixed, as is standard for NHANES Taylor linearization, so it does not reflect variability from the separate nonresponse and calibration adjustments of the two weights being compared.

Item nonresponse was not reweighted. It was small in 2021 to 2023 (25 of 5,754 adults with a phlebotomy weight had no HbA1c result), but in 2017 to March 2020, 451 of 8,451 adults in A (5.3% unweighted; 4.2% weighted) lacked HbA1c under an examination weight that does not account for it. The unweighted share exceeds the 5% level set in the plan.

Pregnancy can be identified only among women aged 20 to 44 in public files, and gestational diabetes history was not collected in 2021 to 2023. The narrow fasting domain was not

reweighted. Group estimates have fewer degrees of freedom, several did not meet the NCHS presentation standards, and no correction was made for multiple comparisons. The splits by diagnosis status and the cut-point analysis were not prespecified and are exploratory.

Conclusions

For NHANES August 2021 to August 2023, and in a separate replication for 2017 to March 2020, HbA1c-defined prevalence in the laboratory domain differed from the paired-domain estimate by an estimated −0.24 (−1.44 to +0.96) and +0.07 (−1.32 to +1.47) percentage points; in 2017 to March 2020 the laboratory domain used the examination weight, which has no blood-draw nonresponse adjustment. The definition difference in the paired domain was −0.41 points (−1.15 to +0.33) and −1.54 points (−2.71 to −0.37), and HbA1c and fasting glucose thresholds classified partly different adults. Comparisons of definitions should use one paired domain and one weight with a covariance-aware difference, and estimates should name their denominator and weight. The accompanying phenotype specification, configuration checker, and aggregate tables support these comparisons.

Abbreviations

AI: artificial intelligence; CI: confidence interval; DOI: digital object identifier; ERB: Ethics Review Board; FPG: fasting plasma glucose; HbA1c: glycated hemoglobin; NCHS: National Center for Health Statistics; NHANES: National Health and Nutrition Examination Survey; PSU: primary sampling unit; SHA-256: Secure Hash Algorithm 256-bit; SQL: Structured Query Language; STROBE: Strengthening the Reporting of Observational Studies in Epidemiology.

Declarations

Ethics approval and consent to participate. This study is a secondary analysis of publicly released, de-identified NHANES data. No new participants were recruited and no participant was contacted. The NCHS Ethics Review Board (ERB) approved the NHANES protocols: Protocol #2021-05 (listed by NCHS as NHANES 2021-2022) for the August 2021 to August 2023 release, whose survey protocols and consent documents NCHS reports as reviewed and approved by the ERB [40], and for the 2017 to March 2020 release the continuation of Protocol #2011-17 through 26 October 2017, Protocol #2018-01 from 26 October 2017, and the continuation of Protocol #2018-01 for 2019 to 2020 [41]. Participants gave informed consent under these protocols; NCHS publishes the consent documents for each cycle [42, 43]. This secondary analysis was conducted in accordance with the Declaration of Helsinki and the NCHS data user agreement [44]. It used only publicly available files without identifiers and therefore did not involve human subjects as defined in the U.S. Common Rule (45 CFR 46.102(e)(1)); no further ethics review or exemption determination was sought.

Consent for publication. Not applicable.

Availability of data and materials. The datasets analyzed during the current study are publicly available from the National Center for Health Statistics: the NHANES August 2021 to August 2023 files DEMO_L, GHB_L, GLU_L, FASTQX_L, DIQ_L, and RHQ_L and the 2017 to March 2020 pre-pandemic files P_DEMO, P_GHB, P_GLU, P_FASTQX, P_DIQ, and P_RHQ, at https://wwwn.cdc.gov/nchs/nhanes/ [16, 17, 21–30, 45]. Their use is subject to the NCHS data user agreement, which permits use for statistical reporting and analysis only and prohibits any attempt to identify participants. Participant-level files are not redistributed with this article.

Additional file 2 contains the source manifest, with the download address and SHA-256 checksum of every file, and the code, SQL, configuration, phenotype specification, and analysis plan; its documented commands download the files from the NCHS pages, verify the checksums, and reproduce every reported estimate. The code and configuration checker are provided under the MIT license. Aggregate results are provided in Additional files 2 and 3. The code is also available at https://anibose11.github.io/projects/diabetes-definition-audit/ [46]. An archived copy of the code (version 1.0.0) is available in the Zenodo repository, https://doi.org/10.5281/zenodo.23450169 [47].

Competing interests. Animesh Bose owns Norlowe Research and has no other competing interests.

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Authors' contributions. Animesh Bose is the sole author and was responsible for conceptualization, methodology, software, validation, formal analysis, investigation, data curation, resources, writing (original draft, review and editing), visualization and project administration (CRediT roles), and read and approved the final manuscript.

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Tables

Table 1. Analysis configurations, sample universes, weights, denominators, and estimates.

Release	ID	Phenotype	Sample universe	Weight	Denominator n	Positive n	Percent (95% CI)	Role	NCHS standard
Aug 2021-Aug 2023	K1	A1C65	P (paired)	WTSAF2YR	2938	371	10.8 (9.1 to 12.7)	primary	presented
Aug 2021-Aug 2023	K2	FPG126	P (paired)	WTSAF2YR	2938	383	11.2 (9.5 to 13.1)	primary	presented
Aug 2021-Aug 2023	K3	A1C65	H (HbA1c laboratory)	WTPH2YR	5729	721	10.5 (9.2 to 12.0)	secondary	presented
Aug 2021-Aug 2023	K4	FPG126	F (fasting glucose)	WTSAF2YR	2940	383	11.2 (9.5 to 13.1)	sensitivity 2	presented
Aug 2021-Aug 2023	K5	DX	P (paired)	WTSAF2YR	2938	409	11.3 (9.3 to 13.5)	composite part	presented
Aug 2021-Aug 2023	K6	TOTAL_A1C_FPG	P (paired)	WTSAF2YR	2938	565	15.8 (13.6 to 18.2)	composite	presented
Aug 2021-Aug 2023	K7	TOTAL_A1C	H (HbA1c laboratory)	WTPH2YR	5729	971	14.0 (12.5 to 15.7)	composite, no glucose	presented
Aug 2021-Aug 2023	K8	BIO_EITHER	P (paired)	WTSAF2YR	2938	470	13.5 (11.4 to 15.8)	sensitivity 3	presented
Aug 2021-Aug 2023	AUX1	TOTAL_A1C	P (paired)	WTSAF2YR	2938	495	13.9 (11.7 to 16.2)	auxiliary	presented
Aug 2021-Aug 2023	K1_S1	A1C65	P-narrow	WTSAF2YR	2766	345	10.4 (8.7 to 12.4)	sensitivity 1	presented
Aug 2021-Aug 2023	K2_S1	FPG126	P-narrow	WTSAF2YR	2766	358	10.9 (9.2 to 12.9)	sensitivity 1	presented
2017-Mar 2020	K1	A1C65	P (paired)	WTSAFPRP	3757	559	10.3 (8.4 to 12.4)	primary	presented
2017-Mar 2020	K2	FPG126	P (paired)	WTSAFPRP	3757	597	11.8 (10.3 to 13.5)	primary	presented
2017-Mar 2020	K3	A1C65	H (HbA1c laboratory)	WTMECPRP	8000	1171	10.4 (9.4 to 11.5)	secondary	presented
2017-Mar 2020	K4	FPG126	F (fasting glucose)	WTSAFPRP	3767	597	11.8 (10.3 to 13.5)	sensitivity 2	presented
2017-Mar 2020	K5	DX	P (paired)	WTSAFPRP	3757	620	11.7 (10.0 to 13.4)	composite part	presented

Release	ID	Phenotype	Sample universe	Weight	Denominator n	Positive n	Percent (95% CI)	Role	NCHS standard
2017-Mar 2020	K6	TOTAL_A1C_FPG	P (paired)	WTSAFPRP	3757	847	16.2 (14.2 to 18.3)	composite	presented
2017-Mar 2020	K7	TOTAL_A1C	H (HbA1c laboratory)	WTMECPRP	8000	1541	14.2 (13.3 to 15.1)	composite, no glucose	presented
2017-Mar 2020	K8	BIO_EITHER	P (paired)	WTSAFPRP	3757	717	14.0 (12.2 to 15.9)	sensitivity 3	presented
2017-Mar 2020	AUX1	TOTAL_A1C	P (paired)	WTSAFPRP	3757	746	14.0 (12.1 to 16.1)	auxiliary	presented
2017-Mar 2020	K1_S1	A1C65	P-narrow	WTSAFPRP	3591	536	10.4 (8.7 to 12.5)	sensitivity 1	presented
2017-Mar 2020	K2_S1	FPG126	P-narrow	WTSAFPRP	3591	572	11.9 (10.4 to 13.5)	sensitivity 1	presented

Weighted percent of the sample universe with 95% Korn-Graubard intervals; denominator n and positive n are unweighted; the last column applies the NCHS Data Presentation Standards for Proportions. K1 to K8 are the eight prespecified configurations; AUX1 is the auxiliary estimate declared for the composite decomposition; K1_S1 and K2_S1 are the sensitivity analysis 1 estimates. Phenotype codes: A1C65, HbA1c ≥ 6.5%; FPG126, fasting plasma glucose ≥ 126 mg/dL; DX, self-reported diagnosed diabetes; BIO_EITHER, either biomarker at or above threshold; TOTAL_A1C_FPG, DX or either biomarker; TOTAL_A1C, DX or A1C65. Domains: A, adults 20+, examined, not pregnant, DIQ010 answered; H, A with an HbA1c result; F, A with a valid fasting glucose; P, F with an HbA1c result; P-narrow, P meeting the narrow fasting rule.

Table 2. Contrasts and decompositions.

Release	ID	Contrast	Left: phenotype; domain, weight; denominator	Right: phenotype; domain, weight; denominator	Difference, percentage points (95% CI)	p
Aug 2021-Aug 2023	C1	HbA1c minus fasting glucose, common paired domain	K1: A1C65; P, WTSAF2YR; n = 2938	K2: FPG126; P, WTSAF2YR; n = 2938	−0.41 (−1.15 to +0.33)	0.25

Release	ID	Contrast	Left: phenotype; domain, weight; denominator	Right: phenotype; domain, weight; denominator	Difference, percentage points (95% CI)	p
Aug 2021-Aug 2023	C2	HbA1c in its laboratory domain minus HbA1c in the paired domain	K3: A1C65; H, WTPH2YR; n = 5729	K1: A1C65; P, WTSAF2YR; n = 2938	−0.24 (−1.44 to +0.96)	0.68
Aug 2021-Aug 2023	C3	Naive comparison: HbA1c laboratory domain minus glucose paired domain	K3: A1C65; H, WTPH2YR; n = 5729	K2: FPG126; P, WTSAF2YR; n = 2938	−0.65 (−2.02 to +0.72)	0.33
Aug 2021-Aug 2023	C4	Glucose in its fasting domain minus glucose in the paired domain	K4: FPG126; F, WTSAF2YR; n = 2940	K2: FPG126; P, WTSAF2YR; n = 2938	−0.01 (−0.02 to 0.00)	0.20
Aug 2021-Aug 2023	C5	Composite without glucose (HbA1c domain) minus full composite (paired domain)	K7: TOTAL_A1C; H, WTPH2YR; n = 5729	K6: TOTAL_A1C_FPG; P, WTSAF2YR; n = 2938	−1.75 (−3.26 to −0.24)	0.03
Aug 2021-Aug 2023	C6	TOTAL_A1C in HbA1c domain minus TOTAL_A1C in paired domain	K7: TOTAL_A1C; H, WTPH2YR; n = 5729	AUX1: TOTAL_A1C; P, WTSAF2YR; n = 2938	+0.17 (−1.31 to +1.66)	0.81
Aug 2021-Aug 2023	C7	TOTAL_A1C minus TOTAL_A1C_FPG, paired domain	AUX1: TOTAL_A1C; P, WTSAF2YR; n = 2938	K6: TOTAL_A1C_FPG; P, WTSAF2YR; n = 2938	−1.92 (−2.48 to −1.37)	< 0.001
Aug 2021-Aug 2023	C8	Composite minus diagnosed (undiagnosed biochemical positivity), paired domain	K6: TOTAL_A1C_FPG; P, WTSAF2YR; n = 2938	K5: DX; P, WTSAF2YR; n = 2938	+4.49 (+3.26 to +5.71)	< 0.001
Aug 2021-Aug 2023	C9	Composite minus biochemical only (diagnosed with both biomarkers below threshold), paired domain	K6: TOTAL_A1C_FPG; P, WTSAF2YR; n = 2938	K8: BIO_EITHER; P, WTSAF2YR; n = 2938	+2.30 (+1.64 to +2.95)	< 0.001
Aug 2021-Aug 2023	C10	HbA1c minus fasting glucose, narrow fasting domain	K1_S1: A1C65; P-narrow, WTSAF2YR; n = 2766	K2_S1: FPG126; P-narrow, WTSAF2YR; n = 2766	−0.50 (−1.32 to +0.33)	0.22
Aug 2021-Aug 2023	C11	HbA1c, narrow fasting domain minus paired domain (sensitivity 1)	K1_S1: A1C65; P-narrow, WTSAF2YR; n = 2766	K1: A1C65; P, WTSAF2YR; n = 2938	−0.33 (−0.76 to +0.10)	0.12
Aug 2021-Aug 2023	C12	Glucose, narrow fasting domain minus paired domain (sensitivity 1)	K2_S1: FPG126; P-narrow, WTSAF2YR; n = 2766	K2: FPG126; P, WTSAF2YR; n = 2938	−0.25 (−0.65 to +0.16)	0.22
2017-Mar 2020	C1	HbA1c minus fasting glucose, common paired domain	K1: A1C65; P, WTSAFPRP; n = 3757	K2: FPG126; P, WTSAFPRP; n = 3757	−1.54 (−2.71 to −0.37)	0.01

Release	ID	Contrast	Left: phenotype; domain, weight; denominator	Right: phenotype; domain, weight; denominator	Difference, percentage points (95% CI)	p
2017-Mar 2020	C2	HbA1c in its laboratory domain minus HbA1c in the paired domain	K3: A1C65; H, WTMECPRP; n = 8000	K1: A1C65; P, WTSAFPRP; n = 3757	+0.07 (−1.32 to +1.47)	0.91
2017-Mar 2020	C3	Naive comparison: HbA1c laboratory domain minus glucose paired domain	K3: A1C65; H, WTMECPRP; n = 8000	K2: FPG126; P, WTSAFPRP; n = 3757	−1.47 (−2.66 to −0.28)	0.02
2017-Mar 2020	C4	Glucose in its fasting domain minus glucose in the paired domain	K4: FPG126; F, WTSAFPRP; n = 3767	K2: FPG126; P, WTSAFPRP; n = 3757	−0.03 (−0.07 to 0.00)	0.06
2017-Mar 2020	C5	Composite without glucose (HbA1c domain) minus full composite (paired domain)	K7: TOTAL_A1C; H, WTMECPRP; n = 8000	K6: TOTAL_A1C_FPG; P, WTSAFPRP; n = 3757	−1.99 (−3.59 to −0.40)	0.02
2017-Mar 2020	C6	TOTAL_A1C in HbA1c domain minus TOTAL_A1C in paired domain	K7: TOTAL_A1C; H, WTMECPRP; n = 8000	AUX1: TOTAL_A1C; P, WTSAFPRP; n = 3757	+0.19 (−1.41 to +1.78)	0.81
2017-Mar 2020	C7	TOTAL_A1C minus TOTAL_A1C_FPG, paired domain	AUX1: TOTAL_A1C; P, WTSAFPRP; n = 3757	K6: TOTAL_A1C_FPG; P, WTSAFPRP; n = 3757	−2.18 (−2.76 to −1.60)	< 0.001
2017-Mar 2020	C8	Composite minus diagnosed (undiagnosed biochemical positivity), paired domain	K6: TOTAL_A1C_FPG; P, WTSAFPRP; n = 3757	K5: DX; P, WTSAFPRP; n = 3757	+4.55 (+3.76 to +5.34)	< 0.001
2017-Mar 2020	C9	Composite minus biochemical only (diagnosed with both biomarkers below threshold), paired domain	K6: TOTAL_A1C_FPG; P, WTSAFPRP; n = 3757	K8: BIO_EITHER; P, WTSAFPRP; n = 3757	+2.25 (+1.72 to +2.77)	< 0.001
2017-Mar 2020	C10	HbA1c minus fasting glucose, narrow fasting domain	K1_S1: A1C65; P-narrow, WTSAFPRP; n = 3591	K2_S1: FPG126; P-narrow, WTSAFPRP; n = 3591	−1.45 (−2.56 to −0.35)	0.01
2017-Mar 2020	C11	HbA1c, narrow fasting domain minus paired domain (sensitivity 1)	K1_S1: A1C65; P-narrow, WTSAFPRP; n = 3591	K1: A1C65; P, WTSAFPRP; n = 3757	+0.14 (−0.09 to +0.37)	0.23
2017-Mar 2020	C12	Glucose, narrow fasting domain minus paired domain (sensitivity 1)	K2_S1: FPG126; P-narrow, WTSAFPRP; n = 3591	K2: FPG126; P, WTSAFPRP; n = 3757	+0.05 (−0.26 to +0.35)	0.76

651 Differences in percentage points with 95% t intervals using design degrees of freedom. Paired definition contrasts use one domain

652 and weight; selection and weighting contrasts hold the phenotype fixed and change the domain (and, for C2 and C6, the weight); naive

contrasts change both and are shown with their decomposition. Each side gives its configuration, phenotype, domain, weight, and unweighted denominator.

Table 3. Agreement between HbA1c and fasting glucose thresholds in the common paired domain.

Release	Measure	Positive n	Percent or kappa (95% CI)
Aug 2021-Aug 2023	HbA1c ≥ 6.5% and glucose ≥ 126 mg/dL	284	8.5 (7.1 to 10.1)
Aug 2021-Aug 2023	HbA1c ≥ 6.5% only	87	2.3 (1.7 to 3.0)
Aug 2021-Aug 2023	Glucose ≥ 126 mg/dL only	99	2.7 (2.0 to 3.6)
Aug 2021-Aug 2023	Both below threshold	2468	86.5 (84.2 to 88.6)
Aug 2021-Aug 2023	Percent agreement		95.0 (93.8 to 96.1)
Aug 2021-Aug 2023	Cohen’s kappa		0.74 (0.70 to 0.79)
Aug 2021-Aug 2023	Share of either-positive adults positive on HbA1c only		17.0 (13.5 to 20.6)
Aug 2021-Aug 2023	Share of either-positive adults positive on glucose only		20.1 (15.8 to 24.4)
Aug 2021-Aug 2023	Sample universe: domain P, weight WTSAF2YR, denominator n = 2938		
2017-Mar 2020	HbA1c ≥ 6.5% and glucose ≥ 126 mg/dL	439	8.2 (6.7 to 9.9)
2017-Mar 2020	HbA1c ≥ 6.5% only	120	2.1 (1.5 to 2.9)
2017-Mar 2020	Glucose ≥ 126 mg/dL only	158	3.7 (2.9 to 4.5)
2017-Mar 2020	Both below threshold	3040	86.0 (84.1 to 87.8)
2017-Mar 2020	Percent agreement		94.2 (93.4 to 95.0)
2017-Mar 2020	Cohen’s kappa		0.71 (0.66 to 0.75)
2017-Mar 2020	Share of either-positive adults positive on HbA1c only		15.2 (11.4 to 19.0)
2017-Mar 2020	Share of either-positive adults positive on glucose only		26.2 (20.0 to 32.5)
2017-Mar 2020	Sample universe: domain P, weight WTSAFPRP, denominator n = 3757		

Weighted percent of adults in domain P with 95% intervals (Korn-Graubard for cells; t intervals for derived measures); positive n is unweighted.

Figures

Figure 1. Fasting subsample status of adults by age, sex, and race and Hispanic origin. Domain A, August 2021 to August 2023 (n = 6,022), weighted with the examination weight WTMEC2YR. Each bar splits adults into those with a valid fasting glucose, those not assigned to the fasting subsample, those assigned without a blood specimen, and those assigned with a fast outside 8 to less than 24 hours. Hatched segments are suppressed under the NCHS presentation standards and are not interpreted. Cell estimates are in Additional file 1, Table S4. Plotted with AI-assisted study code, checked by the author.

Figure 2. HbA1c and fasting glucose threshold prevalence in the same adults with one weight. Domain P: n = 2,938 (August 2021 to August 2023, WTSAF2YR) and 3,757 (2017 to March 2020, WTSAFPRP). Left, weighted prevalence of HbA1c at or above 6.5% and fasting glucose at or above 126 mg/dL with 95% Korn-Graubard intervals. Right, paired difference (HbA1c minus glucose) with 95% t intervals using design degrees of freedom. Plotted with AI-assisted study code, checked by the author.

Figure 3. HbA1c prevalence by laboratory domain and decomposition of the naive difference. Left, HbA1c at or above 6.5% in domain H (WTPH2YR, n = 5,729, in 2021 to 2023; WTMECPRP, n = 8,000, in 2017 to March 2020) and in domain P (fasting subsample weight). Right, the naive difference (HbA1c in H minus glucose in P) equals the definition part (both in P) plus the sample and weight part (HbA1c in H minus HbA1c in P). The sample and weight part is a descriptive comparison. Lines are 95% intervals. Plotted with AI-assisted study code, checked by the author.

Figure 4. Overlap of self-reported diagnosis, HbA1c, and fasting glucose classifications in domain P. August 2021 to August 2023, n = 2,938, WTSAF2YR. Bars are weighted shares of all adults in domain P with 95% Korn-Graubard intervals and unweighted counts. Adults with no diagnosis and both values below threshold (84.2%) are not drawn. Plotted with AI-assisted study code, checked by the author.

Figure 5. Estimates under the three prespecified sensitivity analyses, August 2021 to August 2023. S1, narrow fasting rule (P-narrow, n = 2,766) compared with domain P; S2, single-biomarker domains H and F; S3, diagnosis, biochemical-only, and composite definitions in domain P. Weighted prevalence with 95% Korn-Graubard intervals. Plotted with AI-assisted study code, checked by the author.

Additional files

Additional file 1 (.pdf). **Supplementary tables, figure, and analysis plan amendments.** Tables S1 to S7, Figure S1, and the eight dated amendments to the analysis plan.

Additional file 2 (.zip). **Source code and reproduction materials.** Source code, SQL, R and Python modules, configuration, phenotype specification, analysis plan, source manifest, aggregate result tables, tests with a synthetic fixture, and instructions to rerun.

Additional file 3 (.xlsx). **Aggregate results workbook.** Definitions, thresholds, domains and weights, configurations, participant flow, estimates, contrasts, decompositions, agreement, overlap, diagnosed adults below threshold, missingness by group, paired differences by group, NCHS reconciliation, checker refusals, analyses not prespecified, and validation summary.

Additional file 4 (.pdf). **STROBE checklist.** STROBE checklist for cross-sectional studies, with the location of each item in the manuscript.